Module 19: Graphs & Graph Algorithms

Corrected diagrams — IDs match the question text / CSV order

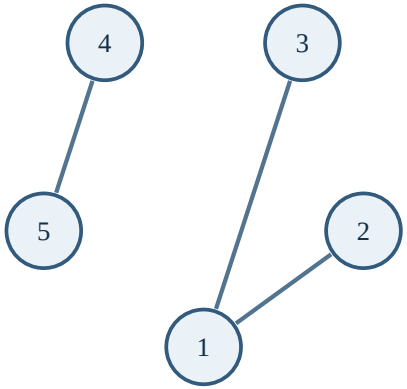
This version keeps the problem-text order: M19-14 Bridges, M19-15 Redundant Connection, M19-16 Bellman–Ford, M19-17 Count Shortest Paths, M19-18 Floyd–Warshall, and M19-19 Grid Shortest Path. Diagram IDs and titles are aligned on every page.

Sample input/output lines are typeset separately. The DFS diagram includes only vertices 1–5; Dijkstra and MST use adjusted layouts to keep weights beside their own edges.

M19-01 — Build an Adjacency List

Difficulty: Easy **Task:** Turn an edge list into a sorted neighbor list for every vertex.

Diagram



Sample input	Sample output
5 3 1 3 1 2 4 5	1: 2 3 2: 1 3: 1 4: 5 5: 4

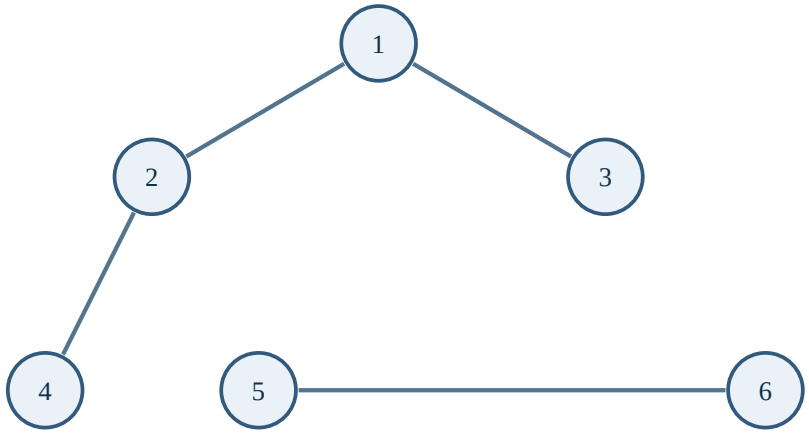
Walkthrough: Vertex 1 lists neighbors 2 and 3 in ascending order. Vertices 4 and 5 form another component.

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M19-02 — Breadth-First Search (BFS) Traversal

Difficulty: Easy **Task:** Visit vertices level by level, starting from source 1.

Diagram



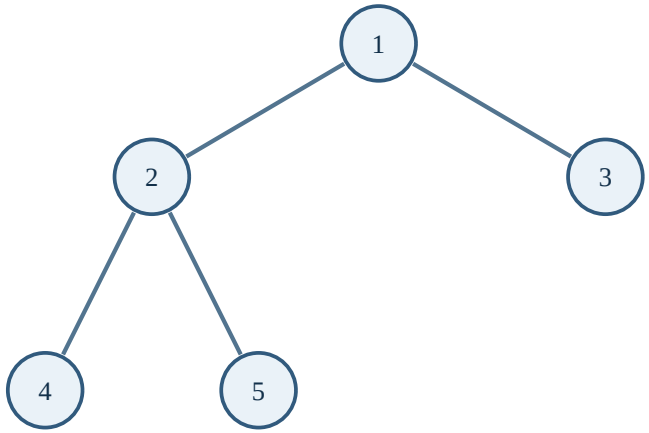
Sample input	Sample output
6 4 1 2 1 3 2 4 5 6 1	1 2 3 4

Walkthrough: BFS visits 1, then 2 and 3, then 4. Vertices 5 and 6 are unreachable from 1.

M19-03 — Depth-First Search (DFS) Traversal

Difficulty: Easy **Task:** Match recursive DFS order with an explicit stack that tracks each vertex's next neighbor.

Diagram



Sample input	Sample output
5 4 1 2 1 3 2 4 2 5 1	1 2 4 5 3

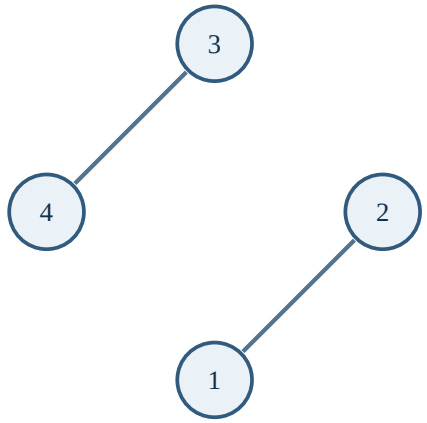
Walkthrough: Visit 1 → 2 → 4, backtrack to 2 and visit 5, then return to 1 and visit 3. Track the next neighbor on the stack.

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M19-04 — Check Whether an Undirected Graph Is Connected

Difficulty: Easy **Task:** Check whether every vertex is reachable from vertex 1.

Diagram



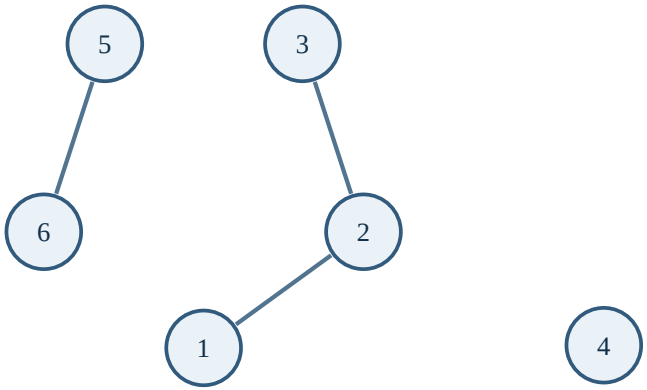
Sample input	Sample output
4 2 1 2 3 4	NO

Walkthrough: Starting at 1 reaches only 1 and 2; vertices 3 and 4 remain unreachable.

M19-05 — Count Connected Components

Difficulty: Easy **Task:** Count disconnected groups; an isolated vertex counts as its own component.

Diagram



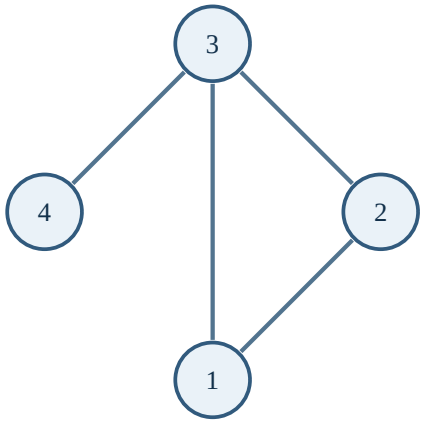
Sample input	Sample output
6 3 1 2 2 3 5 6	3

Walkthrough: The components are {1,2,3}, {4}, and {5,6}. Vertex 4 is isolated and must appear in the diagram.

M19-06 — Detect a Cycle in an Undirected Graph

Difficulty: Medium **Task:** Determine whether an undirected graph contains a cycle.

Diagram



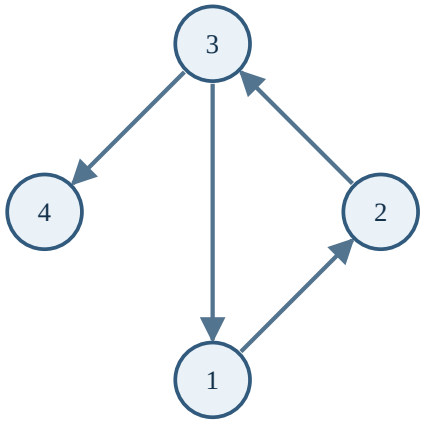
Sample input	Sample output
4 4 1 2 2 3 3 1 3 4	CYCLE

Walkthrough: Edges 1–2, 2–3, and 3–1 form a closed loop. Edge 3–4 is a branch.

M19-07 — Detect a Cycle in a Directed Graph

Difficulty: Medium **Task:** Detect a loop that follows the direction of every edge.

Diagram



Legend: Arrows show edge direction.

Sample input	Sample output
4 4 1 2 2 3 3 1 3 4	CYCLE

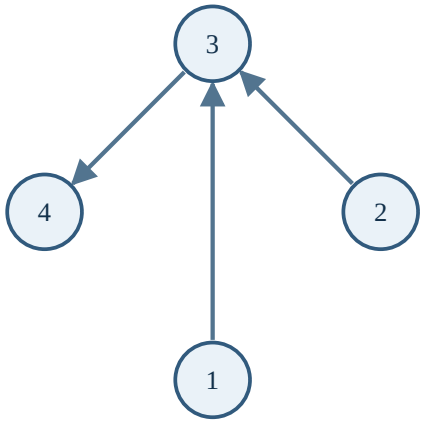
Walkthrough: The directed path 1 → 2 → 3 reaches 1 again, so a directed cycle exists.

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M19-08 — Lexicographically Smallest Topological Order

Difficulty: Medium **Task:** Order tasks so each prerequisite appears before its dependent task.

Diagram



Legend: Arrows show edge direction.

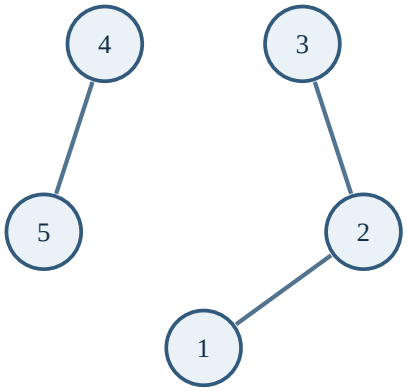
Sample input	Sample output
4 3 1 3 2 3 3 4	1 2 3 4

Walkthrough: Initially 1 and 2 have indegree zero. Choose 1 first, then 2, followed by 3 and 4. A cycle means IMPOSSIBLE.

M19-09 — Shortest Path in an Unweighted Graph

Difficulty: Medium **Task:** Find the minimum number of edges from source 1 to every vertex.

Diagram



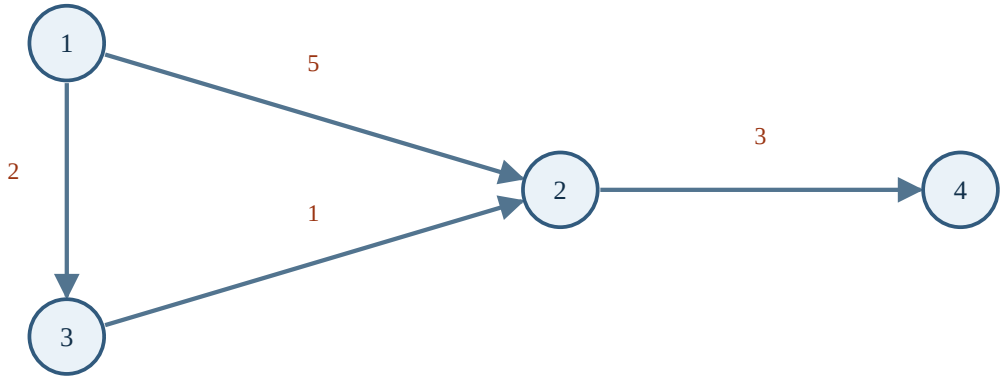
Sample input	Sample output
5 3 1 2 2 3 4 5 1	0 1 2 -1 -1

Walkthrough: Distances from 1 are 0, 1, and 2 for vertices 1, 2, and 3. Vertices 4 and 5 are unreachable.

M19-10 — Dijkstra's Shortest Paths

Difficulty: Medium **Task:** Find minimum total weights from source 1 with nonnegative directed edge weights.

Diagram



Legend: Arrows show edge direction. Numbers are edge weights.

Sample input	Sample output
4 4 1 2 5 1 3 2 3 2 1 2 4 3 1	0 3 2 6

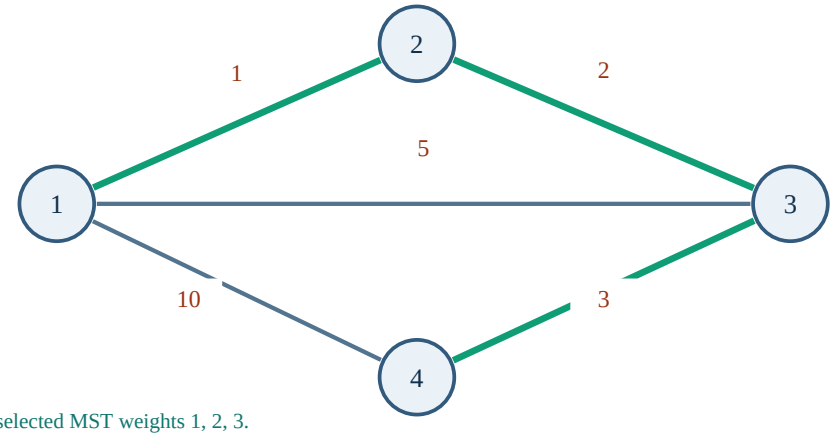
Walkthrough: Route 1 → 3 → 2 costs 2 + 1 = 3, improving direct cost 5. Then reaching 4 costs 3 + 3 = 6.

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M19-11 — Minimum Spanning Tree Weight

Difficulty: Medium **Task:** Connect every vertex with minimum total edge weight and without cycles.

Diagram



Green edges: selected MST weights 1, 2, 3.

Legend: Numbers beside edges are weights/costs.

Sample input	Sample output
4 5 1 2 1 2 3 2 3 4 3 1 4 10 1 3 5	6

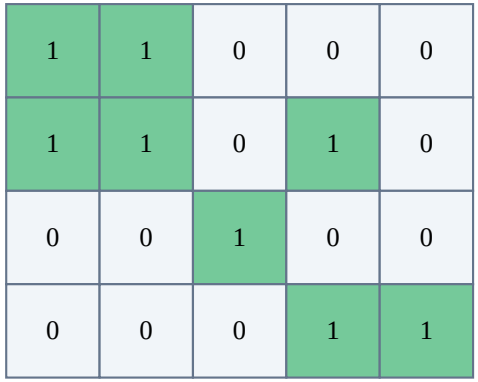
Walkthrough: Select edges with weights 1, 2, and 3. They connect all vertices without a cycle; total weight is 6.

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M19-12 — Count Islands in a Grid

Difficulty: Medium **Task:** Count land groups connected horizontally or vertically; diagonal contact does not connect.

Diagram



Legend: Green = land (1); light = water (0). Diagonal contacts do not connect.

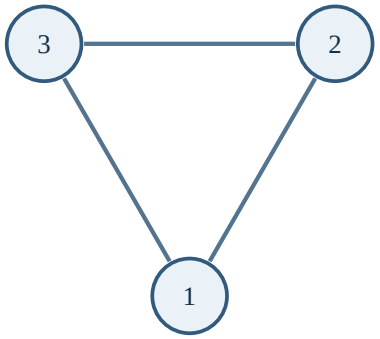
Sample input	Sample output
4 5 11000 11010 00100 00011	4

Walkthrough: Four land groups exist: top-left block, row 2 column 4, row 3 column 3, and the bottom-right pair.

M19-13 — Check Whether an Undirected Graph Is Bipartite

Difficulty: Medium **Task:** Test whether two colors can be assigned so every edge joins different colors.

Diagram



Sample input	Sample output
3 3 1 2 2 3 3 1	NO

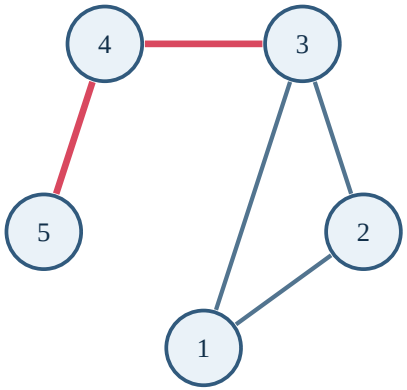
Walkthrough: A triangle forces a color conflict, so two-coloring is impossible.

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M19-14 — Find All Bridges in an Undirected Graph

Difficulty: Hard **Task:** Find edges whose removal increases the number of connected components.

Diagram



Red edges are bridges.

Sample input	Sample output
5 5 1 2 2 3 3 1 3 4 4 5	2 3 4 4 5

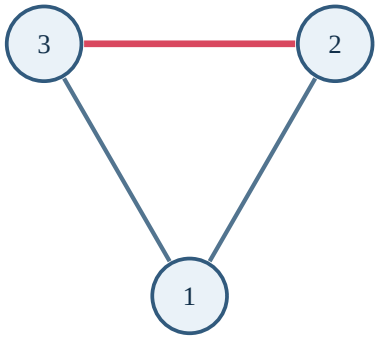
Walkthrough: Removing 3–4 separates the tail from the triangle; removing 4–5 isolates vertex 5. Red edges are bridges.

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M19-15 — Find a Redundant Connection Using DSU

Difficulty: Hard **Task:** Find the first input edge that creates a cycle.

Diagram



Red edge 2–3 is redundant.

Sample input	Sample output
3 1 2 1 3 2 3	2 3

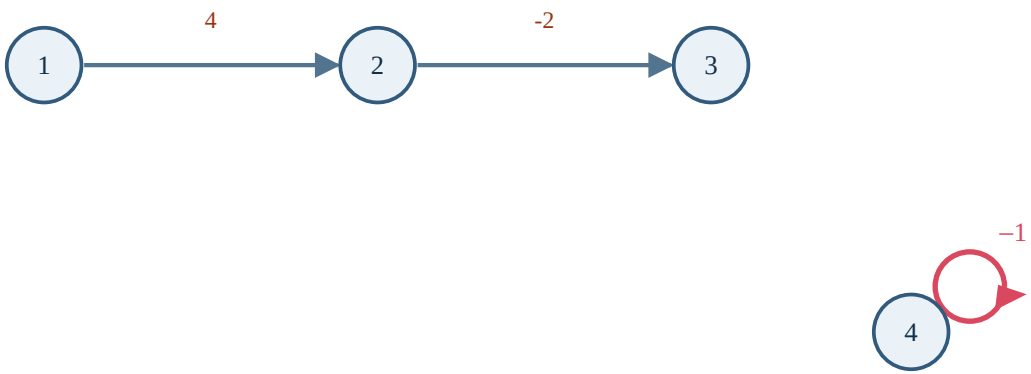
Walkthrough: The first two edges connect 1 to 2 and 3. Edge 2–3 then joins vertices already in the same set; it is highlighted in red.

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M19-16 — Bellman–Ford with Negative Edges

Difficulty: Hard **Task:** Find shortest distances when weights can be negative; only source-reachable negative cycles count.

Diagram



The −1 self-loop is at vertex 4, unreachable from source 1.

Legend: Arrows show edge direction. Numbers are edge weights.

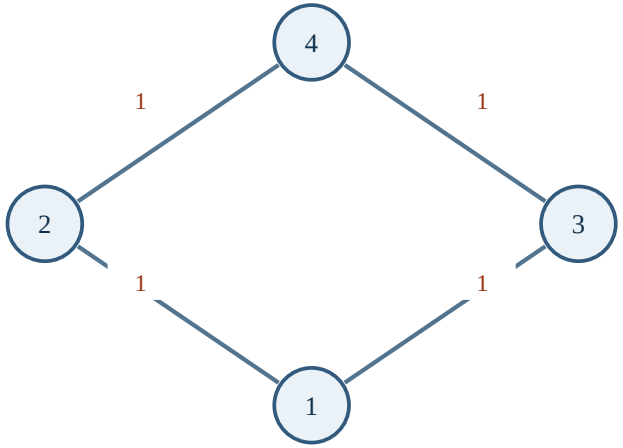
Sample input	Sample output
4 3 1 2 4 2 3 -2 4 4 -1 1	0 4 2 INF

Walkthrough: The negative self-loop at vertex 4 is unreachable from source 1. It is visible but does not change the distances.

M19-17 — Count Shortest Paths Modulo P

Difficulty: Hard **Task:** Count distinct minimum-weight paths from source 1 to target 4.

Diagram



Legend: Numbers beside edges are weights/costs.

Sample input	Sample output
4 4 1 4 1000000007 1 2 1 2 4 1 1 3 1 3 4 1	2

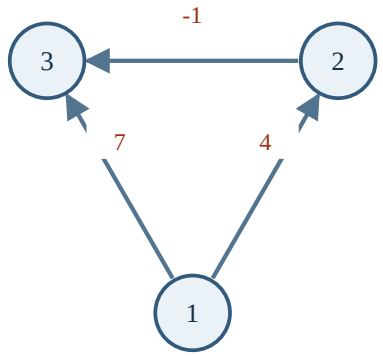
Walkthrough: Two shortest routes have total weight 2: 1 → 2 → 4 and 1 → 3 → 4. All four edge weights are 1.

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M19-18 — All-Pairs Shortest Paths (Floyd–Warshall)

Difficulty: Hard **Task:** Compute shortest distances between every ordered pair in a directed weighted graph.

Diagram



Legend: Arrows show edge direction. Numbers are edge weights.

Sample input	Sample output
3 3 1 2 4 2 3 -1 1 3 7	0 4 3 INF 0 -1 INF INF 0

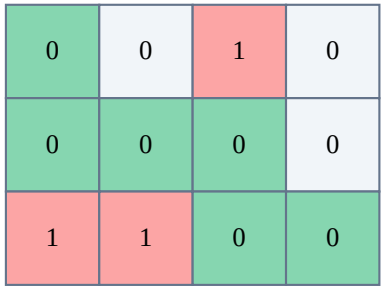
Walkthrough: Route 1 → 2 → 3 costs 3, beating the direct edge of weight 7. INF means no route exists.

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M19-19 — Shortest Path Through a Grid with Obstacles

Difficulty: Medium **Task:** Find the fewest moves from top-left to bottom-right; 0 is open and 1 is blocked.

Diagram



Green cells mark one 5-move shortest route; red cells are blocked.

Legend: Green = cells on one shortest route; red = blocked cells (1); light = other open cells (0).

Sample input	Sample output
3 4 0 0 1 0 0 0 0 0 1 1 0 0	5

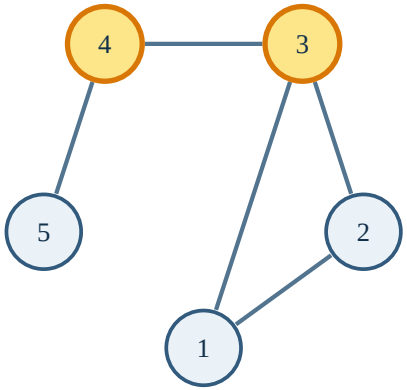
Walkthrough: Highlighted route: (1,1) → (2,1) → (2,2) → (2,3) → (3,3) → (3,4), five moves.

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M19-20 — Find Articulation Points

Difficulty: Hard **Task:** Find vertices whose removal increases the number of connected components.

Diagram



Yellow vertices 3 and 4 are articulation points.

Sample input	Sample output
5 5 1 2 2 3 3 1 3 4 4 5	2 3 4

Walkthrough: Vertices 3 and 4 are highlighted: removing either increases the number of components.

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